

Interview Experience

Pattern Technologies (On-Campus)

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Part 1: The Online Test

It was a MCQ based examination with One Coding Problem. There were 5 Sections :

1. Communication

2. Critical Thinking

Communication and Critical Thinking questions were easy and could be answered with basic common sense.

3. Software Engineer

Software Engineer section consisted objective conceptual questions on Data Structures and Algorithms.

4. Coding: Intermediate-level algorithms

Coding problem named Ciphertext was asked. Similar problem:- <https://www.geeksforgeeks.org/caesar-cipher-in-cryptography/>

Only two test cases were visible for the problem. However, there were more hidden test cases which were difficult to think of. So, after submitting, 40% of my total test cases were passed.

5. MySQL.

MySQL questions were based upon the Database queries. Some common concepts :- Joins, group by, not null;



Part 2: Technical Interview I

Interview started with the 'tell me about what you have done so far' question and I answered it explaining about how my interest was culminated towards software programming, the projects I worked upon and the internship that I did. Next after that, the interviewer asked me, why am I appearing for the support role. Though I was more interested in the software developer's role, I mentioned that I want to explore the things in IT world, and for that, I am ready for the Support role as well. After this point, my actual interview was started.

Question 1:

You have given 3 business entities users, articles, and the comments. Users can post articles and the articles can be commented upon. He clarified that he is not much bothered about who is commenting on the articles. He asked me to design the schema for above business requirement and establish the relations if any.

Question. 2

If you want to find out count of comments on all articles, what query will you write.

To make the things clear, I asked whether there is a user entity involvement in this operation or not and he came back as "let's involve the user as well in this". So, the expected columns in the output were :

user_name | article_name | comments_count

I performed the join on three columns based upon the foreign keys and used group by article_name and user_name.

He gave me the couple of hints and I managed to reach till the correct answer.

He also asked me the reason why we need to have group by clause for all the columns mentioned in the select statement. I tried to explain him the logic behind it.

Question 3.

Third question was the extension of the second question.

Suppose, if we want to find out top 5 users of which the articles have the most number of comments, what query will you go for.

The answer was similar to the answer of the second question with the little changes.

Question 4.

Write algorithm to find out angle between two hands of the clock and the constraints were

1. The time will be given as in 24-hour clock.
2. always need to find the shortest angle.
3. Not to use any fixed formula.

Interviewer was not expecting the movement of hour hand when some minutes are passed which he didn't mention in the question. So, it's important to clarify the doubts regarding the question and to discuss your approach in the initial stages. I was going in a complicated way to solve this question and he brought me on the track and gave me the hint to the approach. I wrote the pseudo code based upon that and then started thinking about the constraints. After solving the complete question, he asked me about the different approaches in his mind and their correctness to solve the problem.

These were the 4 major technical problems that was asked to me in the first technical interview. Apart from this, there were some non-technical questions such as, which technology did you learn by yourself recently? For my surprise, at the end, the interviewer conveyed me that he is forwarding me to an another technical interview for the role of Software Developer.

Part 3: Technical Interview II

It was partially based upon the projects I worked upon, one DSA question and rest were the straightforward conceptual questions.

Project based questions : In my introduction, I mentioned about my previous projects and the interviewer was keen to know more about them. Some of the questions asked which I remember :

1. How did you publish the applications on Play Store? Which build tool did you use?
2. Have you published any cross-platform application?
3. Do you know the platforms and technologies with which we can build the applications that can run on Android as well as IOS with the same code?
4. Why did you consider building native Android Applications instead of the web apps.
5. How does the microservices communicate with each other.
6. CDAC project discussion – what role did you play? What is Authentication? What is Authorization? How security works in Spring boot?

7. Have you heard about web workers?

8. When the large-amount of payload comes from the response, how do you manage it on front-end to avoid the inefficiency in rendering?

- I knew the answer in case of the Android Development which I explained to him.

9. DSA Problem –

Problem Statement. - There is a List<Company<List<Car{

```
Id: "",  
type: "",  
name:"""  
}>>>
```

and you have to convert it to List<type<List<Car>>>

Explanation :- There is a list of Company objects which holds the list of car objects that contain the field called the type of the car. Now, we have to segregate the cars based upon their types and return the list of types of cars.

I managed to solve this problem using Hashmap. Interviewer was expecting the solution using Ecma 6 functions which I was not able to recall.

10. Later on he started touching upon the frontend topics. Some of them were

Promises – what problem does the promise solve, what is the solution to avoid the nesting in promise

Async-await – what are they? How they work?

React features – Declarative Vs Imperative, etc.

11. Do you know about the deadlocks in DBMS?

12. Do you have any questions?

- I asked the interviewer about the job role and his role in the company.

Part 4: HR Interview

1. Introduce yourself.

2. your Family Background.

3. Can you show me your projects which you have mentioned in the resume?

4. Why did you choose IT despite being a chemical engineer.

5. What is your biggest achievement so far?

6. What was your biggest challenge so far?

7. Are you comfortable to work in the US shift?

8. Do you have any question for me?

- I asked about the name 'Pattern'. Does it have any relevance with the philosophy or the goal of the company?

HR was happy with my question as she said she had the same question when she joined the company and she answered me how the person who kept the name to the company answered her!

Preparation -

1. Practicing coding problems on coding platforms helped me to clear the coding rounds.

2. Nilesh Sir's Interview questions discussion sessions and his notes were pretty helpful to clear the concepts for the interviews.

3. Conceptual understanding and ability to explain the part which you have done in your project definitely helps.

4. I had taken up the overview of the part which I haven't implemented in our group project such as Spring Security so that I could have something to speak if the question comes.

All the Best!

For any help regarding the interview preparation, you can contact me at amey36thakur@gmail.com

P.S.

Its okay to have less calls. Its okay not to get shortlisted every time. But don't let it affect your interview preparation. Be confident about yourself, and that one call will make everything well.